

Inverting the Mixture-of-Experts Paradigm: Tiny Reasoning Model Routing of Role-Based Adapters for Small Model Capability Enhancement

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Abstract

We present an early study of tiny reasoning model (TRM) routing over role-based adapters as a path to improving the capabilities of small language models without scaling the dense backbone. Instead of allocating most parameters to experts and learning a gate over them, we invert the usual mixture-of-experts (MoE) pattern: a small base model is paired with multiple role-specialized adapters, while a lightweight reasoning-oriented router selects or conditions which adapter policy should govern a given task. We evaluate several routed lanes against a 4B baseline across 29 common environments spanning reasoning, knowledge, coding, and tool-use-style tasks. The results demonstrate significant capability unlock on low-cost hardware for small models. These findings support a capability-allocation thesis: for small models, task-conditional activation of role-specialized behavior may be more important than simply increasing backbone size. We frame these results as an existence proof and outline a research program for compositional small-model systems built from routed adapters.

1 Introduction

Large-model scaling has dominated recent progress in language model capabilities, but it is not the only way to improve performance [1, 2]. In low-resource and edge-constrained settings, a central question is whether capability can be recovered through better allocation of limited compute and parameters rather than through a larger monolithic network.

This paper explores one such alternative: tiny reasoning model (TRM) routing of role-based adapters. The core idea is to keep the backbone small, train or assemble multiple adapters specialized for distinct behavioral roles, and use a tiny controller to decide which role policy should be active for a given task or context. The TRM has 7 million parameters and takes about 20 minutes to train using 5–6GB of DRAM and a dedicated CPU thread. The resulting architecture treats reasoning and control as a lightweight orchestration problem over specialized behavioral modules. The compact recursive design is consistent with the published TRM architecture, which emphasizes small controller size and strong generalization relative to more elaborate alternatives [7].

This differs from classical MoE in an important way. Standard MoE typically expands the parameter budget by introducing many experts inside the model and training a gate to sparsely activate them [3]. By contrast, our proposal holds the base model small and places specialization largely in external or semi-external adapters. The router is not merely a token-level load balancer but a compact reasoning policy over roles. In that sense, we invert the MoE paradigm: rather than scaling experts and keeping the controller simple, we keep experts lightweight and push more importance onto the controller’s task interpretation and routing choice.

Our hypothesis is that this inversion is especially valuable in the small-model regime, where dense capacity is scarce and indiscriminate scaling yields diminishing returns [1, 2]. If correct, then a well-routed 0.8B or 2B system may outperform a larger plain baseline on meaningful subsets of tasks.

Why would an alignment-focused research program pursue this capabilities research? Alignment and capabilities are not orthogonal, and modeling the near-future of efficient AI ability is essential to understanding the stakes of alignment research for open-source models and evaluating threat vectors. A central question is how much capability compaction can occur using hybrid architectures before dense scaling once again becomes the dominant driver of performance [1, 3, 5].

2 Claim and Contributions

The paper makes four claims.

First, routing policy can dominate parameter count in small-model evaluation. In our shared-environment comparison, routed small-model lanes outperform a 4B baseline on average despite having substantially fewer base parameters.

Second, different routing schemes induce different capability profiles. The generalist-routed lane appears to improve broad benchmark coverage, while the precision TRM lane is more selective but stronger on reasoning-oriented tasks.

Third, adapter routing provides a practical control surface for capability shaping. Rather than retraining a full model to move performance in a desired direction, one can modify the router, adapter inventory, or role decomposition.

Fourth, these findings motivate a broader research agenda in which small-model systems are understood as compositional inference graphs rather than single static policies [5, 6].

Our concrete contributions are:

1. A conceptual framing of TRM-routed role-based adapters as an inversion of MoE for small models.
2. A cross-benchmark comparison over 29 common environments.
3. An empirical observation that routed 0.8B and precision-TRM systems can outperform a plain 4B baseline on average in this setup.
4. A task-family analysis showing that router style appears to trade off broad coverage against concentrated reasoning gains.

3 System Framing

3.1 Architecture intuition

The system consists of three logical components:

- a small base language model,
- a library of role-based adapters representing specialized styles or competencies,
- and a tiny reasoning model router that selects, conditions, or prioritizes the active adapter policy.

The router is intended to operate at the level of task interpretation or behavioral regime selection. Rather than asking a single small model to natively internalize all relevant competencies, routing activates a more appropriate policy for the current problem.

3.2 Why role-based adapters

Role-based adapters are attractive because they permit modular capability injection with lower training cost than full finetuning [4, 5]. They also offer a natural semantic abstraction layer: planner, coder, retrieval synthesizer, strict reasoner, or other task-role decompositions can be represented as composable policies. This is conceptually close to prior work on combining multiple task-specific adapters without destructive interference [6].

3.3 Inverting MoE

In standard MoE, the expert pool is the primary carrier of conditional capacity [3]. Here, the small-model setting motivates a different balance:

- keep the core model compact,
- make specialization cheap and modular through adapters,
- and assign disproportionate importance to the router’s interpretation of the task.

This inversion matters because small systems often fail not only from lack of latent knowledge but from poor behavioral allocation: they use the wrong mode for the task. A compact router that can shift the system into a better role policy may therefore buy more than additional dense parameters alone [1, 3, 6].

4 Experimental Setup

4.1 Compared lanes

We compare a small set of routed and non-routed lanes drawn from the benchmark packs:

- routed 0.8B variants,
- routed 2B variants,
- a routed precision-TRM specialist lane,
- and a plain 4B baseline.

The important comparison is not every intermediate experiment, but the contrast between a larger plain baseline and smaller systems whose behavior is conditioned through router-plus-adapter composition.

4.2 Evaluation slice

We report results on the common-environment slice shared across the compared lanes. This produces a directly comparable environment set and avoids inflating means through non-overlapping benchmark coverage.

4.3 Metric and aggregation

The primary metric is success rate (SR) per environment. Aggregate comparisons use equal weighting across shared environments, along with per-environment deltas and win/loss/tie counts.

4.4 What this section is and is not claiming

This section describes the comparison protocol, not the entire project chronology. The chronology is still useful context for the introduction or discussion, but the methods section should primarily answer: what was compared, on what slice, and with what aggregation rule.

5 Results and Distributional Figures

5.1 Aggregate performance

Across the 29 shared environments, the mean SR values are:

- Old routed 0.8B: **0.195**
- Old routed 2B: **0.184**
- Generalist-routed 0.8B: **0.172**
- Generalist-routed 2B: **0.023**
- Routed precision 2B: **0.115**
- 4B baseline: **0.034**

The headline result is that several routed small-model lanes outperform the plain 4B baseline on average across the shared slice.

5.2 Figure 1: four-way SR histogram

The first histogram is the overview figure for the paper. It shows the per-environment SR distribution for the four key lanes on the common slice: 4B baseline, generalist-routed 0.8B, routed precision TRM, and old routed 2B.

The figure makes three things visually obvious. First, all lanes are sparse and low-sample enough that mass piles up at a few discrete SR values rather than forming smooth curves. Second, the 4B baseline is heavily concentrated near zero, whereas the routed lanes shift more mass into the 0.333 and 0.667 bins. Third, the routed precision-TRM lane and the stronger routed baselines look different not just in mean but in shape: TRM has a narrower, reasoning-skewed profile, while the broader routed lanes spread gains across more environments.

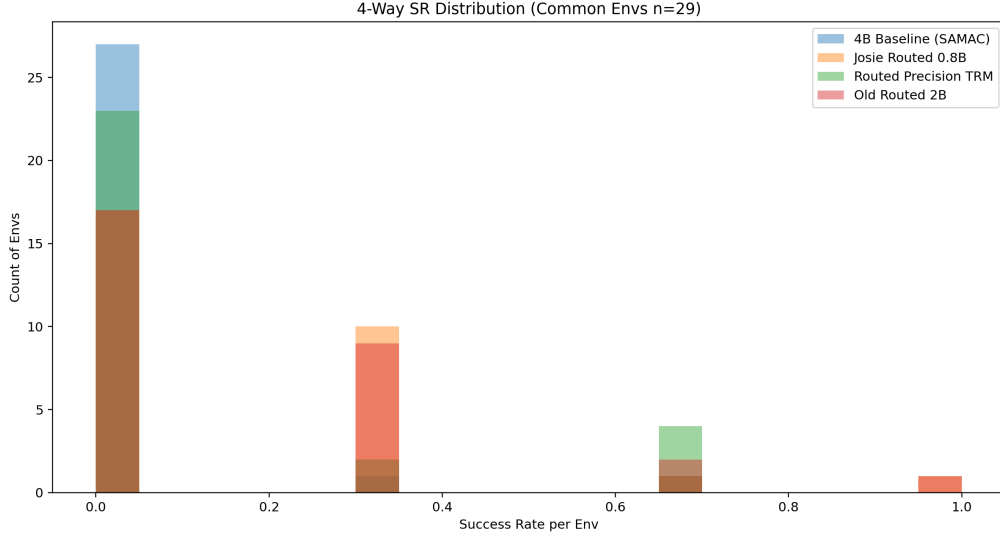


Figure 1: Per-environment SR distribution across four key comparison lanes.

5.3 Figure 2: Generalist minus 4B delta histogram

This histogram supports the broad-coverage claim for generalized routing. The mass is shifted to the positive side, consistent with the generalist-routed lane winning more environments than it loses against the 4B baseline.

The most important interpretation is not merely that the mean delta is positive, but that the distribution contains many non-negative comparisons and several meaningful positive jumps. This supports the view that generalized routing improves broad task adaptation rather than only a tiny handful of environments.

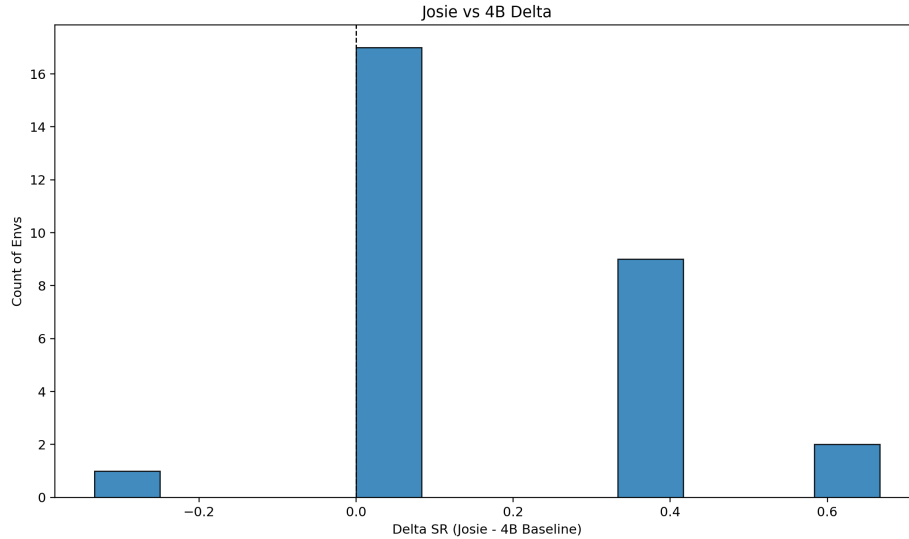


Figure 2: Distribution of per-environment SR deltas for generalist-routed 0.8B minus 4B baseline.

5.4 Figure 3: precision TRM minus 4B delta histogram

This histogram supports a different claim. Precision TRM is also net positive against the 4B baseline, but the pattern is more selective. Instead of broad moderate improvement everywhere, the lane appears to generate concentrated positive movement on a smaller subset of environments, especially reasoning-heavy tasks.

That is exactly the behavior the discussion section should emphasize: precision TRM is not just a weaker generalist. It appears to be a different router regime with a different capability profile.

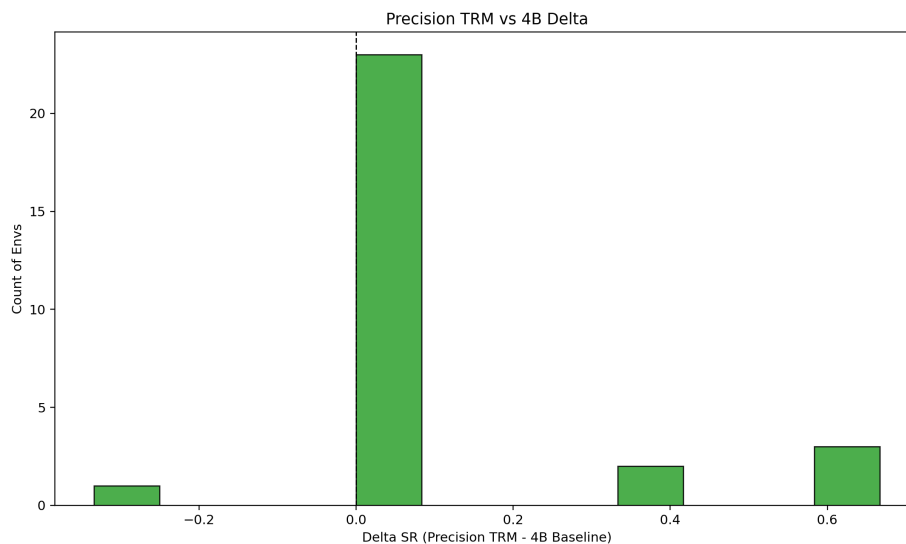


Figure 3: Distribution of per-environment SR deltas for precision TRM minus 4B baseline.

5.5 Figure 4: top and bottom deltas for TRM versus 4B

The bar figure is the task-level interpretability figure. It makes the specialization story concrete by showing where precision TRM most clearly gains and where it gives ground.

This is the figure that best supports the claim that TRM routing is concentrated on reasoning-heavy wins such as ARC Challenge and GSM8K rather than broad coverage across every environment [10, 11].

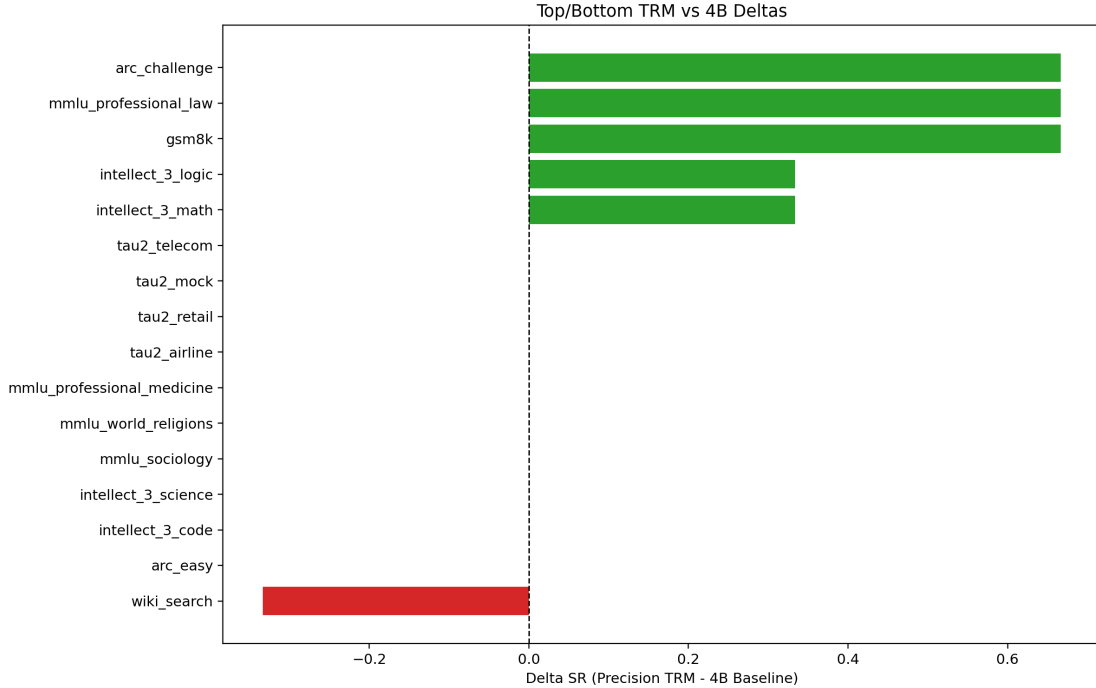


Figure 4: Largest positive and negative per-environment deltas for precision TRM versus the 4B baseline.

5.6 Task-level summary

The figure suite together supports the main empirical story:

- routed small-model systems beat the plain 4B baseline on average in this slice,
- generalized routing looks broader and more coverage-oriented,
- precision TRM looks narrower but stronger on selected reasoning-heavy tasks,
- and the scientific question is therefore not simply whether routing helps, but how router style changes capability shape.

6 Interpretation

6.1 Josified Qwen and the underperformance question

We also tested an ablated Qwen 0.8B model variant, Josified-Qwen, against Qwen 0.8B Base and found that it slightly underperformed in this setting. A natural expectation would be that a more behaviorally shaped model should dominate a simpler routed 0.8B lane. One hypothesis is that the ablation is not the main factor in underperformance but rather that Josified-Qwen is based on the Instruct post-trained version of the model, and that QLoRA adapter training on very small models may have more room to reshape behavior when starting from Base rather than an instruction-tuned checkpoint [9].

The logs make this interpretation more plausible because the issue does not consistently look like routing failure. In many benchmark records, the routed environment matches the target environment

and route-miss counts are zero, yet the Josified-Qwen lane still underperforms on some tasks. That points away from simple misclassification and toward a failure of execution after correct routing: output formatting drift, over-stylized behavior, or an adapter policy that is less legible to the base than a plainer specialization would have been.

This gives the paper a stronger thesis. The lesson is not just that routing works. It is that small-model capability engineering is constrained by compositional geometry. Better controllers and more elaborate policy shaping help only when they pack cleanly into the controllable degrees of freedom of the base model.

A related hypothesis is that the simpler base 0.8B route may have been easier to adapterize robustly. A plain base often presents a cleaner surface for role adapters, whereas a heavily shaped or personality-inflected system may partially spend its limited capacity budget on global style commitments that do not transfer well across many benchmark environments. That would explain why the Josified-Qwen line can still show real wins while not uniformly surpassing the older routed 0.8B setup.

6.2 Variations in Tiny Reasoning Model Training

Using Hierarchical Reasoning Models (HRMs) underperformed TRM, which is consistent with prior research: the published TRM work reports stronger generalization than HRM with a simpler single-network recursive design [7, 8]. However, HRMs may prove uniquely valuable when constellating many TRMs in a more reasoning-model-oriented architecture, such as to orchestrate the use of the TRMs and to apply a mechanistic-interpretability or alignment filter, a control plane to the control planes of the LLM being used.

Additionally, generalizing the number of environments whose datasets were used to train the TRM improved performance relative to specializing them to individual environments. A TRM trained on GSM8K alone underperformed relative to one trained on Intellect-3-Math and GSM8K.

6.3 Methodological cautions

Many environment-level SR values appear in increments of one third, implying small sample counts per environment in the common slice. Aggregate means should therefore be treated cautiously.

The compared lanes are also not a perfect single-variable ablation. Some are older routed systems and some are SAMAC variants. This makes the present paper more of a cross-sectional empirical argument than a fully controlled intervention study.

An additional caution is baseline identity across reports. Some artifacts use local 2B baselines while others use a 4B baseline. A final version should distinguish carefully between within-report comparisons and merged cross-pack comparisons.

7 Why This Matters

There are at least four reasons this line of work matters.

1. It offers a path toward stronger small systems under tight compute or memory budgets.
2. It creates a more interpretable intervention surface. Instead of asking what a monolithic model knows, we can ask which role policy was selected and why.
3. It may reduce iteration cost for research. Adapters and routers can often be modified more cheaply than a full end-to-end model [4, 5].

4. It opens the door to compositional capability engineering, where researchers intentionally build systems out of specialized behavioral modules rather than only scaling a single network [5, 6].

8 Limitations

8.1 Small evaluation counts

Many environment-level SR values appear in increments of one third, implying small sample counts per environment in the common slice. This means aggregate means should be treated as directional evidence rather than a definitive leaderboard claim.

8.2 Incomplete methodological detail

The methods section can be made substantially more concrete in a future revision. The run documentation identifies the key reports, the main training and benchmarking scripts, the runtime split between local Python environments, the precision-TRM specialist training command, and the aggregation logic used to build the paper packs. The logs also support a more narrative methods account, but the current paper still needs a cleaner controlled-ablation presentation.

8.3 Heterogeneous routed systems

The compared lanes are not simply the same model with one variable changed. That makes the present paper more of a cross-sectional empirical argument than a fully controlled ablation.

8.4 Need for significance testing and reruns

The next version should include confidence intervals, bootstrap analysis, or repeated seeded runs to establish whether the observed mean deltas are robust.

9 Research Agenda

9.1 Controlled ablations

One immediate next step is to hold backbone, adapter inventory, benchmark universe, and decoding settings fixed while varying only the router policy and role decomposition.

9.2 Router taxonomy

Explicitly compare broad-coverage routing, precision reasoning routing, retrieval-oriented routing, and hybrid hierarchical routing.

9.3 Multi-step routing

Move beyond single routing decisions to trajectory-aware routing, where the active adapter can change across phases of a problem.

9.4 Storyworld and moral-reasoning environments

Given the broader project context, a particularly interesting direction is to evaluate whether TRM-routed adapters can improve coherent behavior in constrained storyworld and moral tradeoff settings, not just benchmark SR.

9.5 Cost-performance frontier analysis

Quantify capability per unit of parameter count, VRAM, latency, or training cost. The core claim of this architecture is not merely better scores, but better scores for the resource budget.

10 Conclusion

We introduced a framing of tiny reasoning model routing of role-based adapters as an inversion of the mixture-of-experts paradigm for small models. In a 29-environment common slice, several routed small-model lanes outperform a plain 4B baseline on average success rate, and different routing policies yield distinct capability profiles. The generalist-routed lane shows broader wins across benchmark types, while the precision TRM lane shows concentrated gains on reasoning-heavy tasks such as ARC Challenge and GSM8K [10, 11].

These results are preliminary, but they are already strong enough to justify the core research thesis: for small models, capability may be improved substantially by routing specialized behavior more intelligently rather than merely increasing dense model size. This suggests a promising path for small-model research centered on modularity, control, and compositional inference.

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